

Attention-based spatio-temporal-spectral feature learning for subject-specific EEG classification

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Abstract—Brain-computer interface (BCI) is a system that recognizes the human intentions from the brain signals for communication with external devices. The electroencephalography (EEG) signals are commonly used for motor imagery based brain-computer interface (MI-BCI) due to non-invasive, cost-effective, and portable manner. For the analysis of the EEG signals, there are several machine learning and deep learning methods. However, the majority of those methods have limitations of not considering the distinct frequency bands for subject-specific manner. Therefore, we propose the method that pays attention to the significant frequency bands for each subject and also extracts the spatio-temporal-spectral features simultaneously. We utilize filter bank, sliding window segmentation, and the convolutional neural network (CNN) to extract the spatio-temporal features with consideration of multiple frequency bands. Then, we employ the sub-band attention to determine the significant information of each frequency band. Finally, the attention-based Bi-directional Long-Short Term Memory (Bi-LSTM) is implemented to extract the temporal dynamic features. Our proposed method is evaluated on the BCI Competition IV-2a dataset by using two classes in the subject-specific manner. The experimental results demonstrate that our proposed method is effective to focus on the significant frequency band for each subject.

Index Terms—Brain-computer interface, Electroencephalography, Subject-specific manner, Motor imagery, Deep learning, Self-attention, Feature learning, Filter bank.

I. INTRODUCTION

BRAIN-COMPUTER interface (BCI) is a technology that enables people to communicate with external devices by using human brain signals. Specifically, motor imagery based brain-computer interface (MI-BCI) is very useful in clinical applications for control and rehabilitation [1], [2]. The electroencephalography (EEG) signals, which are recorded from a human scalp and capture the brain electrical activities, are commonly used for MI-BCI owing to non-invasive, cost-effective, and portable manner [3]–[5].

For the EEG analysis, three distinctive domains such as spatial, temporal, and spectral domain should be considered [6]–[10]. To extract the spatial features, common spatial patterns (CSP) [7] are widely used in motor imagery classification. Additionally, the sub-band common spatial pattern (SBCSP) [9] and the filter bank common spatial pattern (FBCSP) [8] are introduced to consider the frequency band for each subject. Time domain parameters (TDP) [10] extract the temporal

features of the EEG signals by the test determining the significance of a different number of derivatives of the signals. These methods are built on hand-crafted features and are not an end-to-end approach. Therefore, various methods using deep learning have been proposed to overcome those issues [11]–[13]. However, the majority of deep learning methods still have limitations that they only extract the spatio-temporal features from the broad frequency band [11], [12]. Although few deep learning methods extract the spatio-temporal-spectral features, they do not consider the distinct frequency band for each subject [13].

To address this problem, we propose an attention-based deep learning method, which extracts the spatio-temporal-spectral features considering the significant frequency bands for each subject. Our proposed method consists of three parts: spatio-temporal-spectral feature representation, sub-band attention, and bi-directional attentive temporal dynamics. In the first part, we extract the spatio-temporal features with consideration of multiple frequency bands. In the second part, we utilize the sub-band attention to figure out the significant frequency bands for each subject. The sub-band attention is based on the self-attention module [14] that is effective to focus on the distinct features by allocating attention score to each feature and aggregating those scores. This module has been shown to be successful in various tasks such as computer vision and natural language processing [14], [15]. Therefore, we hypothesize that the sub-band attention could catch the significant frequency band for each subject. In the last part, we implement the attention-based Bi-directional Long-Short Term Memory (Bi-LSTM) [16] to extract the temporal dynamic features by considering distinctive temporal intervals. We employ the components of the convolutional recurrent attention model (CRAM) [12] for our processes of first and last part. We evaluate the performance of our proposed method on the BCI Competition IV-2a dataset [17]. Our result illustrates the improvement of the performance compared to the competing methods. Therefore, the main contribution of our study is to propose the method that is effective to extract spatio-temporal-spectral features in the end-to-end manner and to concentrate on the significant frequency band for each subject, simultaneously.

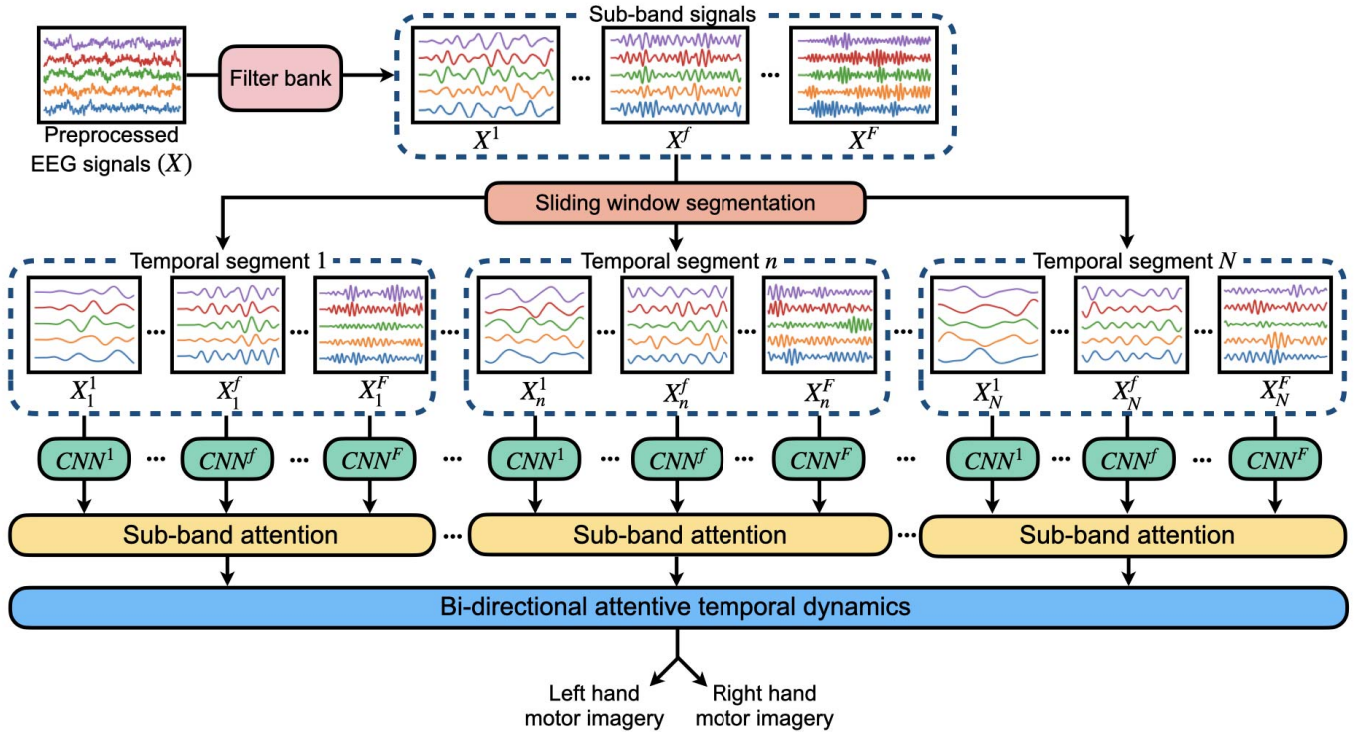


Fig. 1. An overview of our proposed method to extract the spatio-temporal-spectral features for subject-specific EEG motor imagery classification.

II. METHOD

A. Overview

Fig.1 provides an overview of our proposed method for EEG motor imagery classification in the subject-specific manner. There are three major steps in the pipeline. First, the pre-processed EEG signals are separated into the multiple sub-band signals by filter bank. Each sub-band signal is segmented to the temporal segments by sliding window segmentation. The convolutional neural network (CNN) [12], which shares weights depending on the sub-band signals, takes the temporal segments as inputs and generates the spatio-temporal-spectral features as outputs. Second, the spatio-temporal-spectral features from the CNN are fed into the sub-band attention to concentrate on the significant frequency bands. Last, the attention-based Bi-directional Long-Short Term Memory (Bi-LSTM) is applied to the outputs of the sub-band attention in order to extract the dynamic temporal features and determine the importance of each segment. As a result, the softmax classifier generates the prior probabilities in accordance with each class of movement intention.

B. Spatio-Temporal-Spectral Feature Representation

The EEG signals are typically analyzed at three distinctive domains such as spatial, temporal, and spectral domain [6]–[10]. Therefore, we first use filter bank, which contains F frequency bands to consider multiple sub-bands [18]–[20]. The preprocessed EEG signals $\mathbf{X} \in \mathbb{R}^{m \times T}$, where m denotes the number of EEG electrode nodes and T denotes time points, are decomposed into multiple sub-band signals $\mathbf{X}^f$ by filter

bank where f denotes a frequency band. Then, the sub-band signals $\mathbf{X}^f$ are separated into N several temporal segments $\mathbf{X}_n^f \in \mathbb{R}^{m \times w}$ by sliding window technique where n and w denote the temporal segment and the length of the temporal segment, respectively. The CNN is implemented to extract the spatio-temporal features from each temporal segment. To extract the features corresponding to each frequency band, the CNN shares weights depending on the sub-band signals. There are three layers in the CNN sequentially such as a convolutional layer, a max-pooling layer, and a 1×1 convolutional layer. The convolutional layer and the max-pooling layer are applied to squeeze spatio-temporal information and extract important temporal information, respectively. Specifically, the convolutional layer has several filters of which size is $m \times w_c$. These filters are specially designed for capturing characteristics of the EEG signals which represent the connections among different EEG electrodes [12]. The 1×1 convolutional layer is applied to reduce convolutional channels to 1. Following this process, each temporal segment $\mathbf{X}_n^f$ is encoded to the spatio-temporal-spectral features $\{\mathbf{z}_n^f | \mathbf{z}_n^f = \text{CNN}^f(\mathbf{X}_n^f), n = 1, \dots, N, f = 1, \dots, F\}$.

C. Sub-band Attention

The self-attention module is effective to focus on the distinctive features by allocating the attention score to each feature and aggregating these scores [14]. Hence, we apply its self-attention module to pay attention on the distinctive frequency bands and named this module as the sub-band attention. First, the multi-layer perceptrons (MLPs) take the spatio-temporal-

TABLE I
COMPARISON WITH COMPETING METHODS

Comparison Method	Subject									Mean
	S01	S02	S03	S04	S05	S06	S07	S08	S09	
Baseline (0-42 Hz)	0.931±0.007	0.671±0.021	0.942±0.033	0.840±0.023	0.637±0.041	0.783±0.021	0.949±0.016	0.851±0.023	0.962±0.011	0.841±0.117
Delta (0-4 Hz)	0.703±0.040	0.506±0.033	0.514±0.020	0.629±0.039	0.697±0.040	0.635±0.041	0.837±0.026	0.608±0.021	0.620±0.024	0.639±0.100
Theta (4-7 Hz)	0.669±0.032	0.585±0.034	0.674±0.026	0.559±0.029	0.610±0.028	0.634±0.024	0.772±0.022	0.631±0.015	0.655±0.043	0.643±0.064
Alpha (7-13 Hz)	0.965±0.011	0.722±0.022	0.972±0.014	0.749±0.026	0.633±0.022	0.704±0.019	0.792±0.031	0.892±0.019	0.960±0.008	0.821±0.123
Beta (13-30 Hz)	0.966±0.012	0.726±0.018	0.965±0.011	0.863±0.018	0.581±0.026	0.733±0.031	0.978±0.014	0.778±0.017	0.974±0.007	0.840±0.136
Gamma (30-42 Hz)	0.808±0.021	0.663±0.021	0.542±0.026	0.563±0.026	0.547±0.058	0.631±0.045	0.828±0.038	0.660±0.028	0.947±0.007	0.688±0.138
Proposed method	0.963±0.021	0.699±0.033	0.949±0.026	0.840±0.026	0.659±0.064	0.757±0.028	0.980±0.013	0.876±0.032	0.970±0.008	0.855±0.120

spectral features as inputs and generate the attention score $\{\alpha_n^f | \alpha_n^f = MLP(\mathbf{z}_n^f), n = 1, \dots, N, f = 1, \dots, F\}$ where α_n^f indicates the attention score.

D. Bi-directional Attentive Temporal Dynamics

The attention-based Bi-LSTM is employed to extract the attentive dynamic temporal features. The LSTM is effective to handle long-term dependencies between the time points of the EEG signals at a distance. Additionally, to consider bidirectional temporal dynamics, we adopt the Bi-LSTM. A stacked Bi-LSTM takes the outputs of the sub-band attention of all temporal segments as inputs, and returns the hidden states of the second Bi-LSTM layer $\{\mathbf{h}_n \in \mathbb{R}^{l \times 2} | \mathbf{h}_n = Bi-LSTM(\mathbf{w}_n), n = 1, \dots, N\}$, where l indicates the hidden size.

To concentrate on the discriminative temporal intervals, the segment attention is applied on the outputs of the Bi-LSTM. This module operates in the same way as the sub-band attention but only differs in applications:

$$\mathbf{z} = Attention(\mathbf{h}_n), n = 1, \dots, N, \mathbf{z} \in \mathbb{R}^{l \times 2}$$

The output single vector of the attention-based Bi-LSTM $\mathbf{z}$ is fed into the MLP and a softmax classifier sequentially. As a result, a prior probability for each class is generated as $\{\mathbf{p} \in \mathbb{R}^c | \mathbf{p} = softmax(MLP(\mathbf{z}))\}$ where c indicates the number of classes.

III. EXPERIMENTAL RESULTS

A. Dataset and Preprocessing

We evaluate our proposed method on the BCI Competition IV-2a dataset [17]. The dataset contains the EEG signals of four different motor imagery tasks (left hand, right hand, feet, tongue) from 9 healthy subjects acquired in two sessions on different days. To simplify comparison and reduce complexity, we select two motor imagery tasks, *i.e.*, left hand and right hand. Each session consists of 288 trials (72 trials per task) for each subject, and each trial has 6-second length EEG signal with a fixation cross, cue, and motor imagery consecutively. The EEG signals were recorded from the 22 Ag/AgCl electrodes corresponding to the international 10-20 system, sampled with 250 Hz, bandpass filtered between 0.5 Hz and 100 Hz, and notch filtered at 50 Hz to suppress line noise [17].

We use 4.5-second length of each trial from 500 ms before the start cue until the end trial [21]. Thus, each trial includes

the EEG signals which are a two-dimensional (2D) tensor with 22 electrodes and 1125 time points (4.5-second length $\times$ 250 Hz). We also applied exponential moving standardization. Consequently, both training and test sets for each subject have 144 trials.

B. Experimental Settings

The model is implemented with the Pytorch framework. The Adam optimizer with learning rate 10^{-4} is applied to minimize the binary cross-entropy loss function. We use the exponential learning scheduler with learning gamma 0.993 to converge the model. The mini-batch size is 144 which equal to the size of both training and test sets, and the epoch is 400. We determine five frequency bands $F = 5$, *i.e.*, delta (0-4 Hz), theta (4-7 Hz), alpha (7-13 Hz), beta (13-30 Hz), and gamma (30-42 Hz), to perform filter bank [18]–[20]. The sliding window size is 400, and the step size is 50, as a result 15 temporal segments are obtained, *i.e.*, $N = 15$. The convolutional layer has 22×45 size of 40 kernels, *i.e.*, $m = 22, w_c = 45$, and the stride is 1. The max-pooling layer has 1×75 size of a kernel and the stride is 10. The sub-band attention size, the Bi-LSTM hidden size, and the segment attention size are 64, 64, and 512 respectively. We apply the batch normalization after every convolutional layers to get better performance and training speed. Exponential Linear Units (ELU) and Tanh are used as activation functions in the convolutional layers and the attention modules respectively.

C. Results and Analysis

Table I presents a detailed comparison of our proposed method and competing methods. To show the effectiveness of the sub-band attention, we set the baseline method identical to our proposed method, but replace filter bank with bandpass filter between 0 Hz and 42 Hz and remove the sub-band attention. Also, we set several competing methods which are equal to baseline method but only differs in the frequency bands, *i.e.*, delta (0-4 Hz), theta (4-7 Hz), alpha (7-13 Hz), beta (13-30 Hz), and gamma (30-42 Hz). We use accuracy as a criterion for evaluation. However, the accuracy varies from experiment to experiment. To handle this problem, all the procedures of experiments are repeated 10 times, and mean and standard deviation are obtained.

As shown in Table I, our proposed method outperforms the others in terms of mean accuracy. Compared to the baseline method, our proposed method improves about 1.5% mean

TABLE II
ATTENTION SCORE FOR EACH SUB-BAND FOR EACH SUBJECT

Sub-band	Subject								
	S01	S02	S03	S04	S05	S06	S07	S08	S09
Delta (0-4 Hz)	1.764	1.892	1.947	1.898	1.994	1.808	1.887	1.951	1.701
Theta (4-7 Hz)	1.800	1.859	1.844	1.762	1.699	1.867	1.937	2.024	1.764
Alpha (7-13 Hz)	2.152	1.815	2.353	1.841	1.873	1.984	1.891	2.177	2.027
Beta (13-30 Hz)	2.323	2.255	2.184	2.617	1.943	2.606	2.526	1.991	2.218
Gamma (30-42 Hz)	1.962	2.180	1.673	1.882	2.491	1.735	1.759	1.857	2.289

* The orange box indicates the highest attention score for each subject

** The purple box indicates the second-highest attention score for each subject

accuracy and shows best accuracy on all subjects except the subject S04 and the subject S06. Specifically, for the subject 06, it is better to use the broad frequency band, *i.e.*, (0-42 Hz), instead of each sub-band. We believe that our improved accuracy performance is derived from the sub-band attention. As shown in Table II, the orange box and the purple box indicate the highest attention score and the second-highest attention score for each subject, respectively. The sub-band corresponding to the orange box matches to the best sub-band in Table I except for subject S05 and subject S09. Nonetheless, for the subject S05 and subject S09, the purple box matches to the best sub-band in Table I. We demonstrate that our proposed method catches the significant sub-band at each subject.

IV. CONCLUSION

In this work, we propose the method that incorporates the sub-band attention is effective to focus on the significant frequency bands for each subject and to extract the spatio-temporal-spectral features in the end-to-end manner. The experimental results show that our proposed method outperforms the competing methods and demonstrate the effectiveness of the sub-band attention. Thus, our future works can be developing methods for considering more variety of the frequency bands and expanding to the classification of multi-class motor imagery.

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